Unit 3: Technology Roadmap

The challenges for Industry 4.0 transformation are determined as:

- Lack of knowledge about technologies and their opportunities
- Uncertainty about the benefits of technology investments on products and processes
- Lack of knowledge about customer demand regarding new products and busi ness models under industry 4.0 vision
- Limited human and financial resources
- Difficulties to spot the starting point and milestones of planning horizon
- Need for efficient portfolio management for technology investments
- Requirements for prioritization and scheduling of new product and process projects
- Allocating the limited resources to the projects and collaborating with reliable partners
- Lack of communication about the benefits of the Industry 4.0 transformation projects through the organization.

- Therefore, the strategy followed by a company while adopting new technologies to its processes and products has a critical importance. As the first step, the orga nization has to develop a roadmap which is a complex long-term planning instru ment that allows for setting strategic goals and estimating the potential of new technologies, products, and services (Vishnevskiy et al. 2016). The objective of this study is to facilitate to managers, policy makers and practitioners the creation of a technology roadmap for Industry 4.0 transformation. This chapter proposes a comprehensive framework for Industry 4.0 road mapping. The main goal is to overcome the challenges and difficulties confronted by the companies in the digital transformation process.

Introduction

- Technology road mapping is an important method that has become integral to creating and delivering strategy and innovation in many organizations.
- The graphical and collaborative nature of roadmaps supports strategic alignment and dialogue between functions in the firm and between organizations

- **Rob Phaal: The "T-Plan" Pioneer**
  - **The T-Plan (2001):** Phaal introduced this as a fast, low-resource framework for companies to create roadmaps.
  - **Strategic Flexibility:** While originally focused on **market-pull** (customer needs), it became the standard for **technology-push** (new inventions) as well.
  - **The Manager's Tool:** It allows policy makers to quickly link technical development with commercial success.
  - **The Human Factor (2004):** Phaal noted that because many firms lack the right staff, a **qualified specialist** is necessary to manage long-term planning.
- **Daim and Oliver: The Strategic Navigation**
  - **The Travel Metaphor (2008):** They define a roadmap as a "GPS" that identifies the best paths (trajectories) to reach future success.
  - **Three Core Elements:** They argue a roadmap must show the **current state**, the **desired future**, and the **strategy** to bridge the gap.

1. The "To-Do" List for Tech

The process helps a company manage its technology by:

- **Identification:** Finding out what technologies exist.
- **Selection:** Picking the best ones to focus on.
- **Acquisition:** Deciding whether to buy it or build it themselves.
- **Development:** Making the technology work for their specific needs.
- **Exploitation:** Using that technology to actually make money (selling products).
- **Protection:** Keeping it safe from competitors (like using patents).

2. What kind of Tech?

- It isn’t just about gadgets. It includes:
- **Products:** The things customers buy.
- **Processes:** The way the company builds things (faster/cheaper).
- **Infrastructure:** The underlying systems (like software or factories) that keep things running.

3. The Goal

- The reason for doing all of this is to make sure the company stays **stronger than its rivals** (market position) and **makes a profit** (business performance) in a way that fits its **long-term vision** (objectives).

The key benefits of technology road mapping are given as follows:

- Establishing alignment of commercial and technical strategies
- Improving communication across teams and organizations
- Examining potential competitive strategies and ways to implement those strategies
- Efficient time management and planning
- Identifying the gaps between technology, market, and product intelligence
- Prioritizing the investments
- Setting competitive and rational targets
- Guiding and leading the project teams
- Visualizing outputs including goals, processes, and progresses.

1. Aligning Commercial and Technical Strategies

- **The Scenario:** The marketing team sees a "commercial" trend for foldable phones.
- **The Action:** The technical team aligns their R&D (Research and Development) budget to specifically develop flexible glass and hinges so the business goal can be met.

2. Improving Communication Across Teams

- **The Scenario:** Often, engineers and salespeople speak "different languages."
- **The Action:** The roadmap acts as a common visual map so that the **Design team** knows exactly when the **Software team** will have the new OS ready for testing.

3. Examining Competitive Strategies

- **The Scenario:** A competitor announces a phone with 200x zoom.
- **The Action:** The company uses the roadmap to decide if they should implement a similar camera tech or focus on a better battery to win the market

4. **Efficient Time Management and Planning**
   - **The Scenario:** Launching a phone takes 24 months.
   - **The Action:** The roadmap sets "milestones" (deadlines), ensuring the screen is finished by Month 6 so the assembly can start on time in Month 12.

5. Identifying Gaps in Intelligence

- **The Scenario:** The company wants to launch a "satellite calling" feature but realizes they don't have the technical expertise or the partners to do it.
- **The Action:** The roadmap highlights this "gap," prompting the company to either hire new experts or buy a smaller tech firm.

6. Prioritizing Investments

- **The Scenario:** There is only $10 million for R&D.
- **The Action:** Should they spend it on "faster charging" or "better speakers"? The roadmap helps them pick the tech that will bring the most success.

7. Setting Competitive and Rational Targets

- **The Scenario:** The CEO wants a phone that lasts 7 days on one charge.
- **The Action:** The roadmap shows current battery physics. It helps set a "rational" (realistic) target of 2 days, which is still better than the competition.

8. Guiding and Leading Project Teams

- **The Scenario:** A team of 500 engineers is working on different parts of the phone.
- **The Action:** The roadmap gives every lead a clear "North Star," so everyone knows exactly what the final version of the product must look like.

9. Visualizing Outputs and Progress

- **The Scenario:** Shareholders want to know if the company is falling behind.
- **The Action:** The company shows a simplified version of the roadmap—a visual chart showing goals reached and future progress

PROPOSED FRAMEWORK FOR TECHNOLOGY ROADMAP

- Rohrbeck and Schwarz (2013):
  - highlighted "visionary anticipation" of technology is vital for staying competitive in a globalized world.
  - ignoring changes often results in losing opportunities or failing in responding threats.
- Makridakis (2010):
  - role of visioning the future is provide managers and government policy maker’s different ways to comprehend the future and have a whole perception of possible implications of social and technological changes

- “roadmap”:- empowers whoever in the industry to simply understand each move and what decisions need to be done, who needs to make them and when.
- The procedure is decoded into a **project plan** that specifies work across associated stages of formation.
- The proposed framework is designed to help companies visualize future digital transformations and overcome challenges such as lack of technical knowledge, limited resources, and uncertainty regarding investment benefits
- Definition:- A complex long-term planning instrument used to set strategic goals and estimate the potential of new technologies, products, and services.

- The framework is divided into two primary phases: the **Strategy Phase** and the **New Product and Process Development Phase**.
- first phase : the strategies and key technologies are defined
- in the second phase, : development of the new products and processes is conducted.

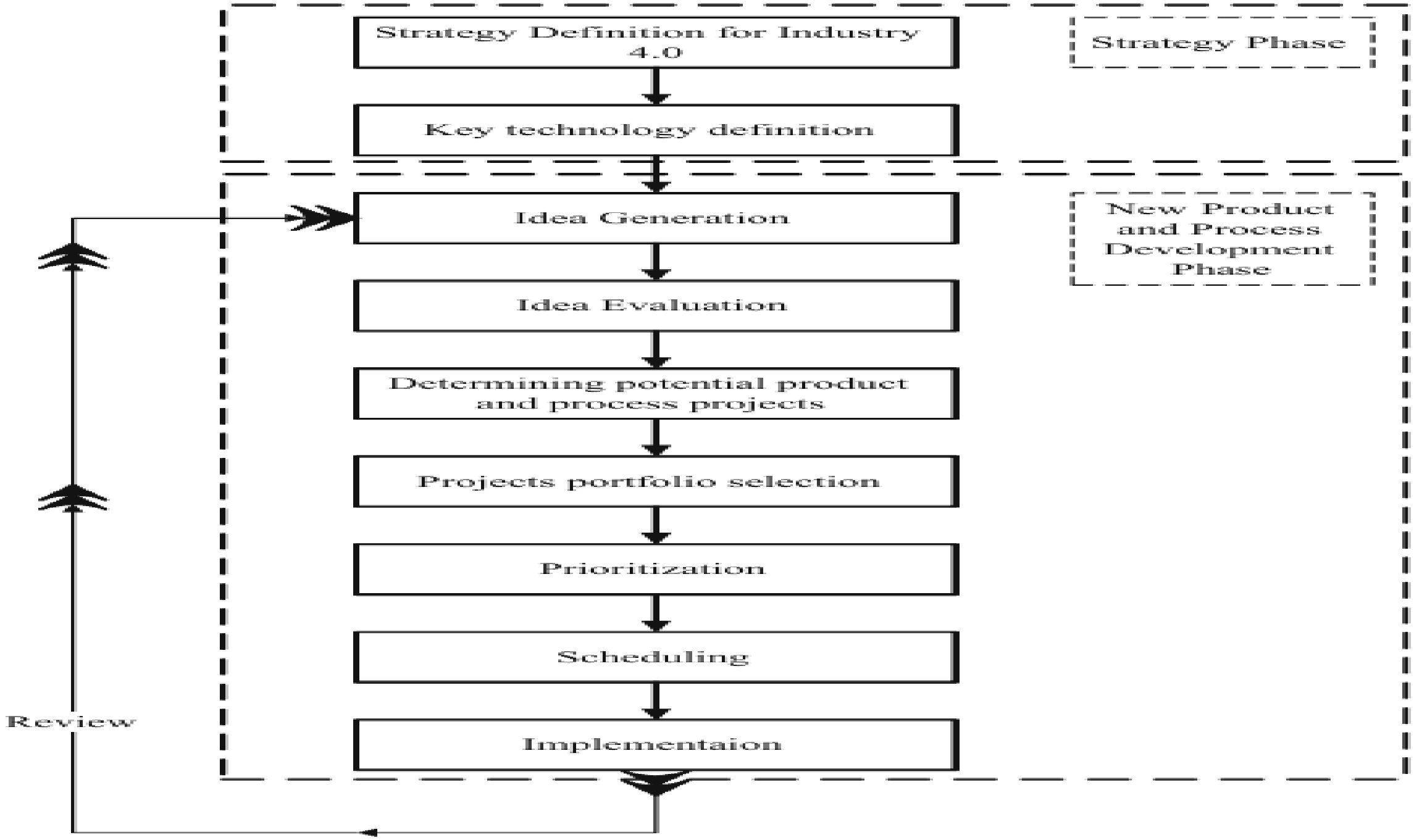

Fig. 5.1 Proposed technology roadmap for Industry 4.0

Strategy Phase

- Strategies answer the question of what has to be done to achieve the desired outcome
- time-based plan and portrays where an industry is, where it needs to go, and how to get it there.
- evaluation of the enterprise digital maturity is done to set clear targets for the next years.
  - help to understand what strengths are possible to build already on, and which systems/processes are needed to combine into future solutions.
- to think about where an organization wants to go in the future, one should consider what the organization could gain by co-operating with **clients, suppliers, technology partners** and even **competitors**, without restricting the vision based on current limitations.

- The focus should be **beyond the technical details** and estimates the impacts of new applications on organization's value chain.
- The roadmap will need to consider future changes in customer behavior and how the organization relationship with them will change.
- Driving from the current to the future desired state will demand precise steps and a clear prioritization
- it is frequently found that technology innovation is a key advantageous factor for performance improvement and survival of the enterprise, besides it is a determinant factor for the sustainable economic growth of nations and quality life improvement of their people

- **Rohrbeck and Schwarz (2013):**
  - blindness caused by focusing mostly on the inside of the company and reinforcing practices that made the company successful in the past thus, it becomes evident that firms need to dedicate efforts to look outside the company and to be aware of the coming changes.
  - the capacity for bringing value to the customer is strongly related to the technology development.

  - Before going to next phase company should have every single detail around the market .

New Product and Process Development Phase

- The new product and process development phase allows sketching goals and projects considering different principles and constraints on separate layers against a shared vision. For our purpose, we use a distinction into three technical perspectives:
- technology constraints (budget and partnerships)
- goals (saving, revenue, risks)
- projects.

- **Project Selection and Portfolio Formation**
- **Expert Evaluation:** Experts select
- 1. generated ideas based on two key dimensions:
- 2. **Product Feasibility(Is it really possible)** and
- 3. **Process Feasibility(Technology we have or not hardware +software)**.example : TIME MACHINE
- 4**. Portfolio Fulfillment:** Once potential projects are listed, they are grouped into portfolios for formal selection.
- **Matrix Flow:** The transition from individual project selection to forming a complete portfolio is managed through a specific flow of matrices.
- NOTE:

•**Portfolio Fulfillment:** Creating a "menu" of potential project combinations for the bosses to choose from.
•**Matrix Flow:** Using a scoring system (grids/charts) to move projects from "just an idea" to "approved for the final list

5. Prioritization Criteria

- The criteria for prioritizing projects differ based on the project type:
- **New Process Development:** Prioritized based on **Savings** vs. **Risk**.
- **New Product Development:** Prioritized based on **Revenue** vs. **Risk**.
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Technology	Budget	Partnership requirment
Technology 1		
Technology 2		
Technology 3		
.		
.		
.		
Technology L		

	Product projects							Process projects						
	Product 1	Product 2	Product 3	.	.	.	Prduct P	Process 1	Process 2	Process 3	.	.	.	Process K
Technology														
Technology 1														
Technology 2														
Technology 3														
.														
.														
.														
Technology L														

	Saving	Revenue	Risk
Product projects			
Product 1			
Product 2			
Product 3			
.			
.			
.			
Product P			
Process projects			
Process 1			
Process 2			
Process 3			
.			
.			
.			
Process K			

Fig. 5.2 Prioritization matrices for product and process projects

At this moment, the generated ideas are selected by experts based on the products and processes feasibility dimensions.

Prioritization stage is conducted considering saving and risk factors

We require optimal portfolio

- optimal portfolio should lie on the Efficient Frontier curve.
- The optimal risk portfolio is normally defined to be around in the center of the curve since as one goes higher up the curve, one takes on proportionately more risk for a lower incremental return.

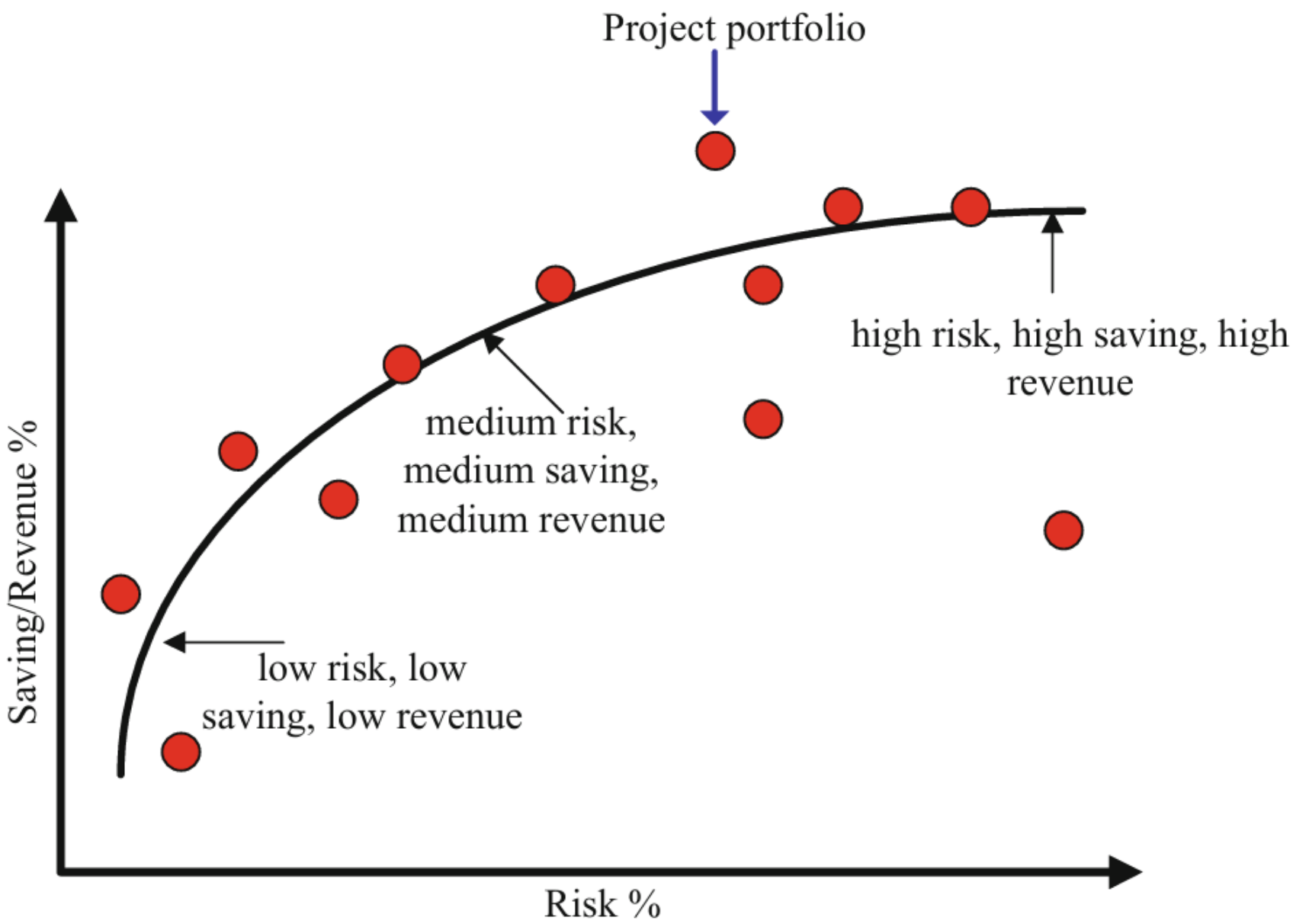

Fig. 5.3 Efficient frontier curve for portfolio selection

- low risk/low return portfolios are pointless since you can gain a similar return by investing in risk-free assets.
- your true selected portfolio will give you the maximum saving and revenue for the percentage of risk you are planning to accept.
- the portfolios above the efficient frontier curve are impossible.
- the portfolios below the efficient frontier are not efficient because for the same risk one could achieve a greater return.

6. Scheduling is creating the **master calendar** for the project. It ensures that people, machines, and money are available at the exact moment they are needed so that the project doesn't stall.

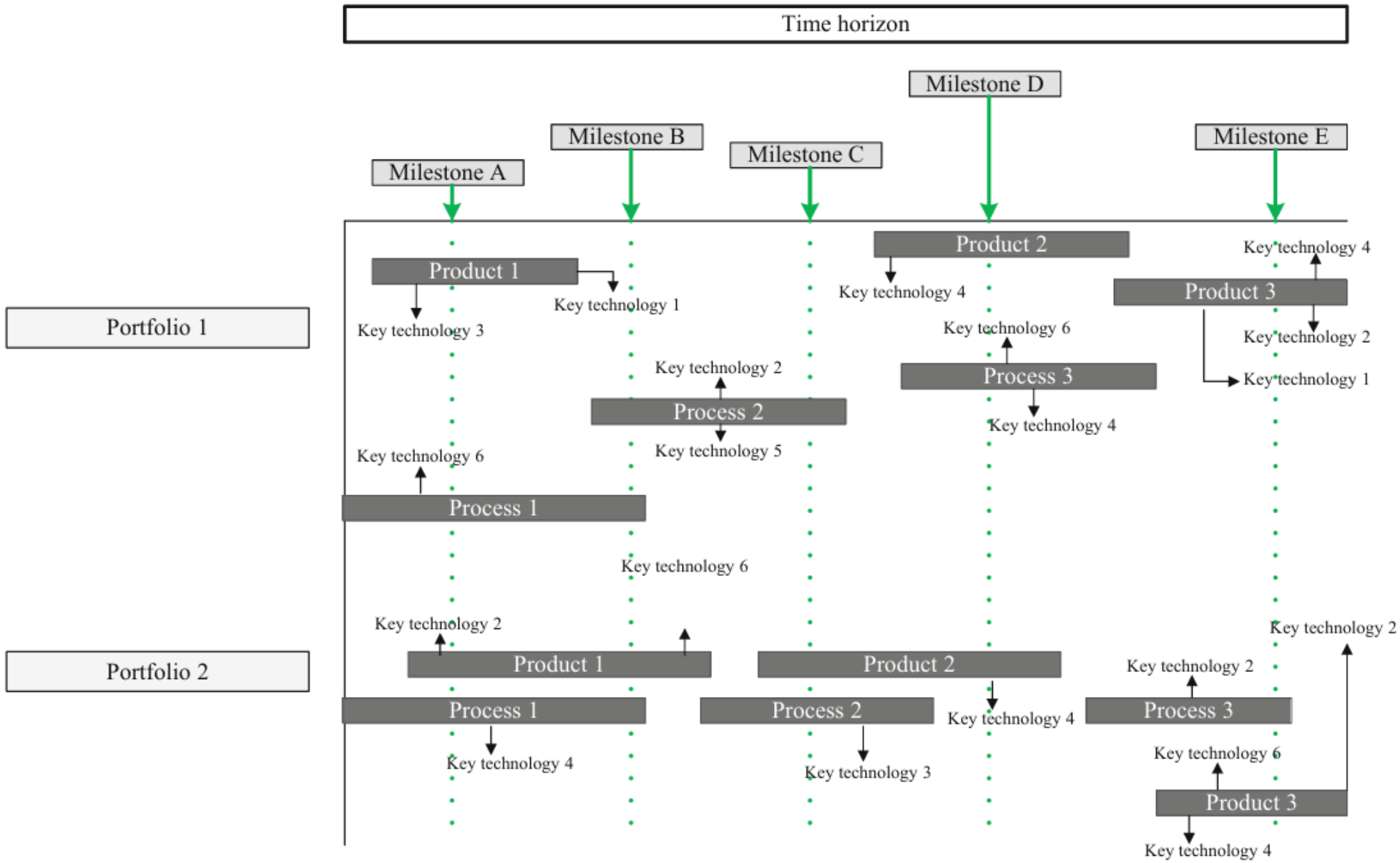

Fig. 5.4 An exemplary scheduled Industry 4.0 roadmap for two portfolios

- Implementation:

- Real coding part ,Deployment part and maintained part work here

- Actually deploy in company .

- To make this clear, let’s use a real-world example: **A traditional furniture factory wanting to reduce waste and speed up production.**

Phase 1: The Strategy Phase

- This is the "brain" of the operation where leadership decides the direction.
- **Strategy Definition for Industry 4.0:** The company decides their main goal is **"Zero-Waste Manufacturing."** They want to move away from manual cutting that wastes wood.
- **Key Technology Definition:** They identify the tools needed. To achieve zero waste, they decide they need **AI-driven cutting algorithms** and **IoT (Internet of Things) sensors** on their saws.

Step	What happens?	Example in the Furniture Factory
Idea Generation	Brainstorming specific solutions.	Ideas: 1) Automated wood sorting, 2) 3D laser scanning of logs, 3) Real-time energy monitoring.
Idea Evaluation	Checking feasibility and cost.	3D laser scanning is found to be highly effective for reducing waste compared to energy monitoring.
Determining Projects	Grouping ideas into formal projects.	The "Smart Cutting Project" is defined, involving hardware (scanners) and software (AI).
Portfolio Selection	Picking the best mix of projects.	Management chooses the "Smart Cutting Project" over a different project to upgrade office computers.
Prioritization	Deciding what comes first.	Rank "Software Integration" as #1 because the hardware can't run without it.
Scheduling	Setting the timeline.	"Phase 1 starts in June; testing begins in August."
Implementation	Doing the work.	Installing the lasers on the factory floor and training the staff.